

Activity - 1

1) Medical Image Enhancement?

Q1: Which gray-level transformation would be recommended?

A1: Log Transformation.

Q2: Why is it suitable?

A2: It enhances the darker region of the x-ray, making soft tissues more visible while keeping bright bone regions clear.

Q3: What effect will it have on the image?

A3: The dark tissues become brighter and more detailed, improve the overall visibility of the x-rays.

2) Night time surveillance?

Q1: Which transformation would improve the visibility?

A1: Log Transformation.

Q2: Why is it more suitable than Negative transformation?

A2: Log transformation increases the brightness of dark pixels whereas negative transforming only inverts the image and does not improve visibility.

Q3: Predict the appearance of the enhanced image?

A3: The image will become brighter, with objects like vehicles, people, and the surroundings appearing more visible and easier to recognize.

3) Finger Print Recognition?

Q1: Which gray level transformation would you apply?

A1: Contrast stretching (piecewise Linear Transformation)

Q2: Explain why?

A2: It increases the contrast between fingerprint, ridges and valleys, making the line ridge details clearer for recognition.

Q3: What would happen if a log transformation were applied instead?
Log Transformation would brighten the image but could not effectively improve the ridge to valley contrast. So fingerprint details may remain unclear.